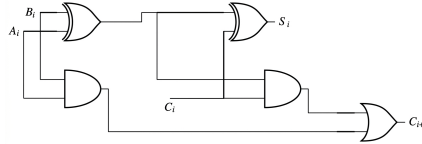


GATE Question Analysis & Hardware Implementation: Full Adder

The GATE Question

Fig.1 shows the i^{th} full-adder block of a binary adder circuit. C_i is the input carry and C_{i+1} is the output carry of the circuit. If the inputs A_i, B_i are available and stable throughout the carry propagation, find the outputs S_i and C_{i+1} .



1 Circuit Analytical Solution

The given logical architecture is a classic standard Full Adder. By tracing the logic gate layout from the inputs, we derive the Boolean behavioral equations:

$$S_i = A_i \oplus B_i \oplus C_i$$
$$C_{i+1} = (A_i \cdot B_i) + (B_i \cdot C_i) + (A_i \cdot C_i)$$

2 Laboratory Components Used

- 1 × Arduino Uno Development Board
- 3 × Tactile Push buttons (Active-LOW style)
- 2 × LEDs (Red: S_i Output, Red: C_{i+1} Output)
- 2 × 220Ω Current-Limiting Resistors
- 1 × Breadboard & M-M Jumper Array

3 Hardware Connection Mapping

- **Inputs:** $A_i \rightarrow$ Pin 2, $B_i \rightarrow$ Pin 3, $C_i \rightarrow$ Pin 4. All pins are bridged to **GND** via buttons using internal pull-up logic.
- **Outputs:** $S_i \rightarrow$ Pin 8 $\rightarrow$ 220Ω $\rightarrow$ Red LED. $C_{i+1} \rightarrow$ Pin 9 $\rightarrow$ 220Ω $\rightarrow$ Red LED.

4 System Truth Table

A_i	B_i	C_i	Sum (S_i)	Carry (C_{i+1})
0	0	0	0	0
1	0	0	1	0
0	1	0	1	0
1	1	0	0	1
0	0	1	1	0
1	0	1	0	1
0	1	1	0	1
1	1	1	1	1

5 Arduino Embedded C++ Code

The firmware code for the digital logical execution environment implemented on the ATmega328P microcontroller can be viewed below:

https://github.com/Anshita9/internship/blob/main/gate_question.ino

6 AVR Assembly Firmware Code

The low-level direct register manipulation and instruction optimization variant for hardware tracking is hosted below:

https://github.com/Anshita9/internship/blob/main/gate_question.asm